□□□□ Chemistry Paper □□□□□□

□□ You are provided with□
- solution **A** aqueous copper □□ sulphate□
- solid **B** iron powder□
- 0.02 M acidified potassium manganate (VII), solution **C**□

□ You are required to determine the molar heat of displacement of copper by iron□

□ **Procedure**□

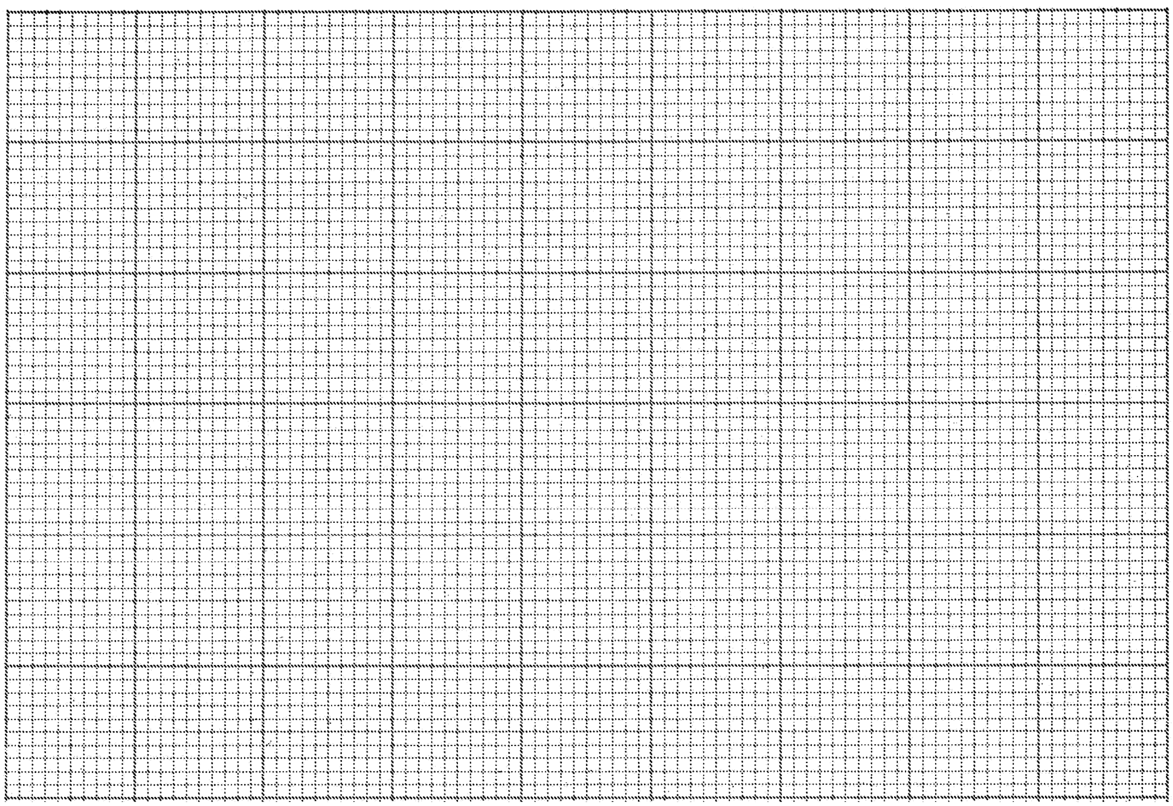
□ Using a burette place □□□□ cm□ of solution **A** in a □□□□ ml beaker□ Measure the temperature of the solution and record it in table □ below□ Add **all** of solid **B** provided at once and start a stop watch□ Stir the mixture **thoroughly** with the thermometer and record the temperature of the mixture after every one minute in the table□ **Retain** the mixture for use in procedure □ below□

□ **Table**□

Time □Min□	□	□	□	□	□	□	□	□
Temperature □C□								

□□marks□

□ □a□ □i□ Plot a graph of temperature □vertical axis□ against time in the grid provided□
□□marks□



□ □ □ii□ From the graph determine the□

(I) highest change in temperature, ΔT; (1 mark)

(ii) time taken for reaction to be completed. ($\frac{1}{2}$ mark)

(iii) Calculate the heat change for the reaction. (Specific heat capacity of solution is [illegible] $Jg^{-1}K^{-1}$; Density of the solution is 1 gcm^{-3}) ([illegible] marks)

Procedure II

Carefully decant the mixture obtained in procedure I into a 250 ml volumetric flask. Add about [illegible] cm^3 of distilled water to the residue in the [illegible] ml beaker. Shake well, allow the mixture to settle and carefully decant into the volumetric flask. **Immediately** add about [illegible] cm^3 of 2 M sulphuric (VI) acid to the mixture in the volumetric flask. Add more distilled water to make [illegible] cm^3 of solution. Label this as solution **D**.

Fill a burette with solution **C**. Using a pipette and a **pipette filler** place [illegible] cm^3 of solution **D** into a 250 ml conical flask. Titrate solution **D** against solution **C** until the **first permanent pink** colour is obtained. Record your results in table [illegible] below. Repeat the titration two more times and complete the table. Retain the remaining solution **C** for use in question [illegible].

Table [illegible]

	I	II	III
Final burette reading			
Initial burette reading			
Volume of solution C used (cm^3)			

([illegible] marks)

(a) Determine the average volume of solution **C** used. ($\frac{1}{2}$ mark)

(b) Calculate the number of moles of:

(i) aqueous potassium manganate (VII) used. ([illegible] mark)

(ii) iron (II) ions in [illegible] cm^3 of solution **D** ([illegible] mole of MnO_4^- reacts with [illegible] moles of Fe^{2+}) ([illegible] mark

(iii) iron (II) ions in [illegible] cm^3 of solution **D**. ([illegible] mark)

(c) Calculate the molar heat of displacement of copper by iron. ([illegible] marks)

[illegible] You are provided with solid **E**. Carry out the following tests and write your observations and inferences in the spaces provided.

(a) Place **all** of solid **E** in a boiling tube. Add about [illegible] cm^3 of distilled water and shake thoroughly. Filter the mixture into another boiling tube. **Retain** the filtrate for use in test 2(b) below. Dry the residue using pieces of filter papers.

(i) Transfer about half of the dry residue into a dry test-tube. Heat the residue strongly and test any gas produced using a burning splint.

Observations	Inferences
([illegible] mark)	([illegible] mark)

(ii) Place the rest of the residue in a dry test tube. Add [illegible] cm^3 of [illegible] M hydrochloric acid. **Retain** the mixture for test (iii) below.

Observations	Inferences
([illegible] mark)	([illegible] mark)

(iii) To [illegible] cm^3 of the solution obtained in (ii) above, add [illegible] cm^3 of aqueous ammonia dropwise.

Observations	Inferences
([illegible] mark)	([illegible] mark)

(b) (i) To [illegible] cm^3 of the filtrate obtained in (a) above, add about 3 cm^3 of aqueous ammonia (Excess).

Observations	Inferences
([illegible] mark)	([illegible] mark)

(ii) To [illegible] cm^3 of the filtrate, add about 2 cm^3 of [illegible] M hydrochloric acid.

Observations	Inferences
([illegible] mark)	([illegible] mark)

(iii) To [illegible] cm^3 of the filtrate, add one or two drops of barium nitrate solution.

Observations	Inferences
([illegible] mark)	([illegible] mark)

[illegible]. You are provided with solid **G**. Carry out the tests in (a) and (b) and write your observations and inferences in the spaces provided. Describe the method used in part (c).

(a) Place about one third of solid **G** on a **metallic** spatula and burn it in a Bunsen burner flame.

Observations	Inferences
([illegible] mark)	([illegible] mark)

(b) Dissolve all of the remaining solid **G** in about [illegible] cm^3 of distilled water in a boiling tube. Use the solution for tests (b) (i), (ii) and (c).

(i) Place [illegible] cm^3 of the solution in a test-tube and add 2 drops of acidified potassium manganate (VII) solution.

Observations	Inferences
([illegible] mark)	([illegible] mark)

(ii) To [illegible] cm^3 of the solution, add all of solid sodium hydrogen carbonate provided.

Observations	Inferences
([illegible] mark)	([illegible] mark)

(c) Determine the pH of the solution obtained in (b) above.

Method used	Inferences
([illegible] marks)	([illegible] mark)